Optimization

Optimization is when we want to find the maximum and minimum values of the function.

Coin toss

Product rule

$$\frac{dg}{dp} = \frac{d}{dp}(p^7(1-p)^3) = \frac{d(p^7)}{dp}(1-p)^3 + p^7\boxed{\frac{d\left((1-p)^3\right)}{dp}}$$

Chain rule

$$= 7p^6(1-p)^3 + p^7\boxed{3(1-p)^2(-1)}$$

$$= p^6(1-p)^2\left[7(1-p) - 3p\right]$$

$$= \boxed{p^6}\boxed{(1-p)^2}\boxed{(7-10p)} = 0$$

$$\cancel{p = 0} \qquad \cancel{p = 1} \qquad p = 0.7$$

An alternative and faster way

$$\log(g(p)) = \log(p^7(1-p)^3) = \log\left(p^7\right) + \log\left((1-p)^3\right)$$

$$= 7\log(p) + 3\log(1-p) = G(p)$$

$$\frac{dG(p)}{dp} = \frac{d}{dp}(7\log(p) + 3\log(1-p)) = 7\frac{1}{p} + 3\frac{1}{1-p}(-1)$$

$$= \frac{7(1-p) - 3p}{p(1-p)} = 0$$

$$7(1-p) - 3p = 0 \qquad p = 0.7$$

Why we choose the logarithm instead?

1. Derivative of products is hard, derivative of sums is easy

$$f(p) = p^6(1-p)^2(3-p)^9(p-4)^{13}(10-p)^{500}$$

$$\frac{df}{dp}$$

$$\begin{aligned}
&[6p^5](1-p)^2(3-p)^9(p-4)^{13}(10-p)^{500}+\\
&p^6\left[2(1-p)\right](3-p)^9(p-4)^{13}(10-p)^{500}(-1)+\\
&p^6(1-p)^2\left[9(3-p)^8\right](p-4)^{13}(10-p)^{500}(-1)+\\
&p^6(1-p)^2(3-p)^9\left[13(p-4)^{12}\right](10-p)^{500}+\\
&p^6(1-p)^2(3-p)^9(p-4)^{13}\left[500(10-p)^{499}\right](-1)
\end{aligned}$$

$$\frac{d}{dp}\log(f)$$

$$\begin{aligned}
&\frac{6}{p}+\frac{2}{1-p}(-1)+\frac{9}{3-p}(-1)+\\
&\frac{13}{p-4}+\frac{500}{10-p}(-1)
\end{aligned}$$